

Introduction

This report presents a Login and Registration System developed in C++. The program allows users to register with a username and password, and then log in using those credentials. Credentials are stored securely in a text file.

Features

1. User Registration: Users can register with a unique username and password.
2. Input Validation: Username must be at least 3 characters, password at least 4 characters.
3. Duplicate Check: The system checks if a username already exists.
4. Password Confirmation: Users must confirm their password during registration.
5. Secure Storage: Credentials are stored in users.txt in the format username|password.
6. Login Verification: Users can log in with their credentials.

Code Overview

- usernameExists() — Checks if a username already exists.
- registerUser() — Handles user registration.
- loginUser() — Handles user login.
- File Handling: Uses ifstream and ofstream to read/write credentials.

Output

```
=====
CodeAlpha LOGIN & REGISTRATION SYSTEM
=====

1. Register
2. Login
3. Exit

-----
Enter your choice (1-3): 1

-----
REGISTRATION
-----
Enter username: IQRA MAQSOOD MUGHAL
[!] Username already exists. Please choose another.

Press Enter to continue...
```

Figure 1: Login System Output

```
=====
CodeAlpha LOGIN & REGISTRATION SYSTEM
=====

1. Register
2. Login
3. Exit
-----
Enter your choice (1-3): 1
-----

REGISTRATION
-----
Enter username: Iqra Maqsood Mughal
Enter password: 123456
Confirm password: 123456

[OK] Registration successful! You can now log in.

Press Enter to continue..._
```

Figure :Registration System Output

Conclusion

The Login and Registration System demonstrates file handling, string manipulation, and input validation in C++. It provides a basic but functional authentication system.

References

1. C++ Documentation. (2026). *File Input/Output*. Retrieved from <https://en.cppreference.com>
2. Stroustrup, B. (2018). *The C++ Programming Language*. Pearson.